The Price of Time Travel: Three Information-Theoretic Barriers from Minimum Fisher Information

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Abstract. We show that the gravitational entropy production $\Sigma = -\ln(-g_{00})$, derived from channel extensivity, satisfies the vacuum equation $\nabla^2\Sigma = 0$. The maximum principle for this harmonic field establishes three barriers to exotic spacetime structures, each with a quantifiable "price" to overcome: (1) event horizons require stellar-mass energy concentration—a price Nature already pays; (2) warp bubbles require $E \propto v^4 R^2/\delta$ of localized energy—enormously expensive but finite; (3) closed timelike curves require $\Sigma < 0$, meaning regions where time runs faster than at infinity—the most expensive barrier, requiring physics beyond the harmonic approximation. We compute each price explicitly and identify experimental signatures at the boundary of Barrier 1 testable by the Event Horizon Telescope.

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I. The Idea

How much does a time machine cost?

Not in dollars—in physics. Every science-fiction device that manipulates spacetime—warp drives, wormholes, time machines—violates some property of the gravitational field. But violations have prices. This essay calculates them.

The starting point is an observation Einstein made in 1911, four years before general relativity [1]: the coordinate speed of light slows down near a massive body, $c_{\text{eff}} = c\sqrt{-g_{00}}$. Dicke reformulated this as a refractive-medium picture in 1957 [2]. We define

$$\Sigma \; = \; -\ln(-g_{00})\,, \tag{1}$$

and call it the *information cost* of gravity. This is not an ad hoc definition. In quantum information theory, the gravitational redshift acts as a pure-loss bosonic channel with transmissivity $\eta = -g_{00}$ [3]. If a photon traverses two gravitational regions in succession, the transmissivities multiply: $\eta_{\text{tot}} = \eta_1 \cdot \eta_2$. Any extensive measure of information loss must then satisfy $\Sigma(\eta_1\eta_2) = \Sigma(\eta_1) + \Sigma(\eta_2)$—the Cauchy functional equation, whose unique continuous solution is $\Sigma = -\ln\eta$ [4]. Channel extensivity *forces* equation (1).

What field equation does Σ satisfy in vacuum? The principle of minimum information gradient says: minimize the Fisher information—the Cramér–Rao lower bound on how precisely the mass distribution can be inferred from the field— $\mathcal{F}[\Sigma] = \int |\nabla\Sigma|^2\, d^3x$. The Euler–Lagrange equation is the Laplace equation, $\nabla^2\Sigma = 0$. The spherically symmetric solution with $\Sigma \to 0$ at infinity and the correct Newtonian limit is $\Sigma = r_s/r$, giving the exponential metric $g_{00} = -e^{-r_s/r}$ [5, 6].

Here is the key insight. The *maximum principle* for harmonic functions—that Σ cannot attain a local extremum in the interior of the domain—simultaneously forbids three exotic spacetime structures. Each forbidden structure can be "purchased" by supplying enough energy to make $\nabla^2\Sigma \neq 0$, and the price is calculable.

Gravity, viewed through the Σ lens, is spacetime's price list. The maximum principle is the budget constraint. This essay reads the receipt.

II. The Maximum Principle

The engine of everything that follows is a classical theorem from the theory of partial differential equations [7]. We state it in the form we need:

Theorem (Strong Maximum Principle). *Let $\nabla^2\Sigma = 0$ in a connected domain D, with $\Sigma \to 0$ at infinity. Then:*

(i) $\Sigma > 0$ *everywhere in D (no negative values);*

(ii) $\sup(\Sigma)$ *is attained only on ∂D (no interior maximum);*

(iii) Σ *is uniquely determined by boundary conditions.*

The proof fits in two lines of any PDE textbook. But its consequences for gravity are profound. Property (i) says $g_{00} = -e^{-\Sigma}$ never crosses -1: time always runs *slower* than at infinity. Property (ii) says Σ cannot pile up at a point: no infinite redshift surface. Property (iii) says the gravitational field is rigid, completely fixed by what happens at the boundary.

Gravity, rewritten in the Σ language, inherits all the rigidity of harmonic analysis. The three barriers to exotic spacetimes are three corollaries of (i)–(iii). And every violation has a price, because the only way past a harmonic barrier is to break harmonicity—which costs energy.

III. Barrier 1: Event Horizons

An event horizon requires $g_{00} \to 0$ at some finite radius, which means $\Sigma \to \infty$. But property (ii) of the maximum principle forbids any interior extremum of a harmonic function, let alone a divergence. The harmonic solution $\Sigma = r_\mathrm{s}/r$ gives $g_{00} = -e^{-r_\mathrm{s}/r}$, which is strictly negative for all $r > 0$. There is no horizon.

To *get* a horizon, one must break harmonicity: $\nabla^2\Sigma \neq 0$, which requires a source. In the presence of matter, Σ satisfies a Poisson equation $\nabla^2\Sigma = 8\pi G\rho/c^2$, and concentrating a mass M into a radius $r \leq r_\mathrm{s} = 2GM/c^2$ creates a region where Σ is driven past all bounds by the matter source. The Schwarzschild metric emerges when ρ is compressed to a point.

The price of Barrier 1: concentrate mass M into $r \leq 2GM/c^2$.

Nature already pays this price—but only inside matter. The vacuum field equation $\nabla^2\Sigma = 0$ applies outside the star; inside, the Poisson equation $\nabla^2\Sigma = 8\pi G\rho/c^2$ takes over. The horizon forms at the boundary of the matter distribution during collapse, where Σ transitions from the Poisson regime to the harmonic regime. The open question is: what is the vacuum extension beyond the collapsing matter? Schwarzschild (with horizon) or exponential (without)? The 4.6% shadow difference is the empirical arbiter.

Every stellar-mass black hole is a purchased violation of the maximum principle, financed by gravitational collapse. Every supermassive black hole at a galactic center is a receipt for that transaction.

But here is the surprise. The "cheapest" solution—the one with $\nabla^2\Sigma = 0$ everywhere, no source, no horizon—predicts a *different* photon sphere. In isotropic coordinates, the exponential metric's effective potential for photons, $V_{\text{eff}} = e^{-2r_s/\rho}/\rho^2$, has its maximum at $\rho_{\text{ph}} = r_s$, giving a critical impact parameter $b_c = e \cdot r_s \approx 2.72\, r_s$, compared with the Schwarzschild value $b_c = 3\sqrt{3}\, r_s/2 \approx 2.60\, r_s$. The difference is 4.6%.

The Event Horizon Telescope's measurement of the M87* shadow currently has ∼10% uncertainty [8]. The next-generation EHT, with additional baselines and shorter wavelengths, aims for ∼1% precision. At that level, Barrier 1 becomes directly testable: does Nature pay the *full* price (Schwarzschild, with horizon), or only the *minimum* price (exponential, without)?

IV. Barrier 2: Warp Bubbles

A warp bubble is a region of modified spacetime that moves at velocity v while the interior remains flat. In the Σ language, this requires $\Sigma = 0$ in a flat interior, $\Sigma > 0$ in a thin shell of thickness δ, and $\Sigma = 0$ in the exterior. That is: a local maximum of Σ confined to a shell of radius R.

Property (ii) of the maximum principle forbids this. If Σ is harmonic, it cannot have a local maximum in the interior.

To purchase a warp bubble, one must again break harmonicity. The required energy density in the shell sources $\nabla^2\Sigma \neq 0$ and can be estimated by dimensional analysis from the field equation. The Alcubierre warp metric [9] in the lapse sector gives a shell amplitude

$\Sigma_0 \sim v^2/c^2$, and the integrated energy is

$$E_{\rm warp} \sim \frac{c^4}{8G}\frac{v^4}{c^4}\frac{R^2}{\delta}\,, \tag{2}$$

where R is the bubble radius and δ the wall thickness.

The v^4 scaling is the key. Subluminal warp is exponentially cheaper than superluminal warp.

Speed	Energy	Mass equivalent
c (light speed)	$\sim 10^{28}\,\mathrm{kg}$	3 Jupiter masses
$0.01\,c$	$\sim 10^{20}\,\mathrm{kg}$	large asteroid
$100\,\mathrm{m/s}$	$\sim 67\,\mathrm{kg}$	one human
$1\,\mathrm{m/s}$	$\sim 60\,\mathrm{MJ}$	1.5 liters of gasoline

Table 1: Price list for a warp bubble of radius $R = 10\,\mathrm{m}$ and wall thickness $\delta = 1\,\mathrm{m}$. The v^4 scaling makes subluminal warp extraordinarily cheap—in principle.

The price of a walking-speed warp bubble is 1.5 liters of gasoline. The catch: in the lapse formulation, it must be *negative* energy, shaped into a specific Σ profile. Nobody knows how to do this.

A crucial caveat: Fuchs *et al.* [10] have shown that subluminal warp does *not* require negative energy when formulated via the shift vector rather than the lapse. Our Σ framework is purely a lapse-sector analysis, so it overestimates the difficulty: the true price may be lower than equation (2) suggests. Whether the lapse and shift formulations can be reconciled within the information-theoretic framework is an open question.

What is *not* an open question is the scaling. Whatever the sign of the required energy, the v^4 dependence means that near-light-speed travel is vastly more expensive than gentle subluminal nudges. The universe's price list has a steep luxury tax on speed.

V. Barrier 3: Closed Timelike Curves

Now we come to time travel itself.

A closed timelike curve (CTC) requires, among other conditions, a region where $g_{00} < -1$, which means $\Sigma < 0$: time runs *faster* than at infinity. This is a *necessary* condition for CTCs—the spacetime must also have the right causal topology to close a timelike path—but the maximum principle forbids even this first step. Property (i) states that $\Sigma > 0$ everywhere for a harmonic field with $\Sigma \to 0$ at infinity. Without $\Sigma < 0$, no CTC can form. Barrier 3 is qualitatively different from Barriers 1 and 2. Those required large Σ (strong gravity); this requires Σ of the *wrong sign.*

The price of Barrier 3: a "negative information source," meaning $\nabla^2\Sigma < 0$—a region that *accelerates* time rather than slowing it.

What does $\Sigma < 0$ mean physically? At $\Sigma > 0$, clocks tick slower than at infinity (normal gravity; the GPS correction). At $\Sigma = 0$, flat space, normal time. At $\Sigma < 0$, clocks tick *faster* than at infinity. Push Σ negative enough, past a threshold, and the light-cone tips over: a CTC forms.

Does $\Sigma < 0$ exist anywhere in known physics?

The Kerr black hole.—In the ergosphere of a spinning black hole, $g_{00} > 0$: the time coordinate becomes spacelike. This is actually *worse* than $\Sigma < 0$—it is a signature change where $\Sigma = -\ln(-g_{00})$ is not even defined. The Penrose process extracts energy from the ergosphere, but does not directly create a CTC.

The Gödel universe.—Gödel's 1949 rotating cosmology [11] has $g_{00} < -1$ in certain regions, and it famously contains CTCs. But it requires the *entire universe* to rotate uniformly. The price is a global constraint: a specific stress-energy tensor ($\rho + p > 0$ with uniform rotation) pervading the entire universe.

The Alcubierre–CTC connection.—In special relativity, faster-than-light travel implies time travel: two superluminal trips with a Lorentz boost between them close into a CTC. So a superluminal warp drive (Barrier 2 at $v > c$) automatically triggers Barrier 3. The two barriers are *independent* for subluminal warp, but *linked* for superluminal warp. This is why Hawking conjectured the chronology protection principle [12]: Nature may enforce a cosmic censorship not only of singularities but of time machines.

In the Σ framework, chronology protection is not a conjecture but a theorem—*within the harmonic approximation*. The maximum principle *proves* that $\Sigma < 0$ is impossible if $\nabla^2\Sigma = 0$. The question is whether the universe allows sources that violate harmonicity in the required direction.

The combined price of a time machine:

1. Overcome Barrier 2: supply $\sim 10^{28}$ kg of energy in a thin shell (for light-speed warp).
2. Overcome Barrier 3: create a $\Sigma < 0$ region—currently unknown physics.
3. These are *independent* barriers. Both must be paid.

A time machine is the most expensive object in the universe. Its price tag has two parts: an engineering challenge (Barrier 2, enormous but finite) and a physics challenge (Barrier 3, currently unknown). We have computed the first; the second remains the province of future theory.

VI. The Price List

We can now read the universe's receipt. Table 2 collects all three barriers, their physical meaning, what breaks them, and the cost.

Barrier	Forbidden	What breaks it	Price	Status
1. Horizon	$\Sigma \to \infty$ at finite r	Mass concentration	$M > c^2 r/(2G)$	Already paid
2. Warp	Interior max of Σ	Energy source ($\nabla^2\Sigma \neq 0$)	$E \sim c^4 v^4 R^2/(8Gc^4\delta)$	Finite but vast
3. CTC	$\Sigma < 0$	Negative info source ($\nabla^2\Sigma < 0$)	Unknown	New physics

Table 2: The price list. All three barriers follow from a single theorem—the maximum principle for the harmonic information field Σ. Gravity is the universe's price list for manipulating time.

Remarkably, all three prices emerge from a single two-line theorem. The maximum principle for harmonic functions, applied to the gravitational information field $\Sigma = -\ln(-g_{00})$, gives the complete tariff schedule. Property (i) forbids time machines. Property (ii) forbids warp bubbles and horizons. Property (iii) guarantees that the cheapest solution—the exponential metric—is unique.

The exponential metric itself, $g_{00} = -e^{-r_s/r}$, is the "zero-price" gravitational field: the solution that satisfies the vacuum equation everywhere, with minimum Fisher information, and no exotic structures. It passes all solar-system tests through second post-Newtonian order ($\beta = \gamma = 1$ in the PPN framework [17]), with deviations from Schwarzschild appearing only at $O((r_s/r)^3) \sim 10^{-18}$ for the Sun—far below the Cassini bound [18]. But in the strong field, near compact objects, the 4.6% shadow difference becomes accessible.

The connection to prior work runs deep. Jacobson [16] showed in 1995 that Einstein's equation is an equation of state; Dorau and Much [15] derived it from the quantum relative entropy in 2025. Our contribution is to show that the *same* information-theoretic structure that gives the field equation also gives the *barriers*—the maximum principle is not an add-on but a built-in feature of the information field. The barriers are the field equation's way of saying "no."

VII. Outlook

Three observations close this essay.

First, the exponential metric—the harmonic, horizonless, minimum-Fisher-information solution—is testable now. The ngEHT will reach the $\sim$1% shadow precision needed to distinguish $2.72\, r_{\mathrm{s}}$ from $2.60\, r_{\mathrm{s}}$. If the exponential metric is correct, there is no event horizon, and the black hole information paradox takes on a new character—not requiring information to escape from behind a horizon, because the horizon was never there. The most expensive item on the price list (Barrier 1) turns out to be the one Nature may not have purchased after all.

Second, the v^4 scaling of warp energy makes subluminal warp tantalisingly cheap in magnitude but frustratingly expensive in kind (negative energy, in the lapse formulation). The recent positive-energy warp solutions of Fuchs *et al.* [10] suggest that the shift sector may offer a way around the negative-energy requirement. In the Σ framework, this would correspond to a generalization of the harmonic condition from the lapse to the full metric—a direction we are actively exploring.

Third, the most profound lesson is negative. The maximum principle does not merely make time travel expensive; it makes it *categorically* different from the other two barriers. Horizons and warp bubbles require large Σ—a quantitative challenge. CTCs require $\Sigma < 0$—a qualitative one, demanding sources of a type not known to exist. This is not a no-go theorem—it is a *price-go* theorem. It says exactly what must be paid, and for Barrier 3, the currency has not yet been invented.

The question, then, is not whether time travel is possible. The maximum principle tells us it is forbidden in the harmonic approximation. The question is: what is the price of going beyond harmonic? And can the universe afford it?

Gravity is spacetime's price list. The maximum principle is the budget constraint. And time travel is the item we cannot yet afford.

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